Fpt University

TRƯỜNG ĐẠI HỌC FPT

CAPSTONE PROJECT REPORT

Report 5 – Software Test Documentation

– Ho Chi Minh City, October 2026 –

Table of Contents

I. Record of Changes 3

II. Testing Documentation 4

1. Scope of Testing 4

2. Test Strategy 5

2.1 Testing Types 5

2.2 Test Levels 6

2.3 Supporting Tools 6

3. Test Plan 7

3.1 Human Resources 7

3.2 Test Environment 7

3.3 Test Milestones 8

4. Test Cases 8

5. Test Reports 9

I. Record of Changes

Date	A* M, D	In charge	Change Description
25/10/2026	A	Nguyễn Phương Mỹ Thuận	Initial test documentation for Review 2: scope, strategy, test plan, 75 functional test cases (Report5_Test_Report.xlsx), 51 unit test cases (Report5_Unit_Test.xlsx) and round 1 results on tag review-2.

*A - Added M - Modified D - Deleted

II. Testing Documentation

1. Scope of Testing

The target of test is the Review 2 release of OptiPack AI (repository tag review-2, commit 1a3101c): the NestJS back end, the OptiPack staff portal (React), the AURELLE website (Next.js) and the optional packer service (Python, OR-Tools CP-SAT).

In scope. The test-case set follows the formula of the Review 2 guide (section 9.3) and the targets in the Design Brief, Appendix A3:

Source	Items	Minimum per item	Test cases
Mandatory features F-01 … F-10 (Design Brief section 6)	10	3 (main flow; data or business rule; exception or permission)	30
Main workflows WF-01 … WF-06 (end-to-end, main scenario)	6	1	6
State transitions of the Order Group (allowed-status-transitions.ts)	33	1	33
Agreed NFRs NFR-1 … NFR-6 (Design Brief section 7)	6	1	6
Total			**75**

Out of scope (not tested in this release): the Flutter mobile app (planned for Iteration 3), order synchronization for Tiki, TikTok Shop, Shopee and Facebook Marketplace, write-back to Lazada, live carriers other than GHN, Excel / PDF analytical report export, and the home-dashboard widgets that still show sample data (Design Brief, Known Gaps G1, G2).

Test levels

Level	In charge	Inputs / time	Focus	Acceptance criteria
Unit	Developer of each module (Thuận, Thi for back end)	Source code and Jest specs; on every commit and push	Business rules of services, solvers, state transitions; MongoDB models mocked	All Jest tests pass in CI (job "Backend (typecheck, lint, test)")
Integration	Thuận (marketplace, GHN), Thi (packer)	AURELLE Open API simulator, GHN staging API, Lazada test shop; each iteration	Webhooks, signatures, write-back, carrier booking and status, packer service	Conformance / smoke scripts pass; no duplicated events
System	Whole team, each workflow by its owner	Staging (or local Docker Compose) with seeded data; Verification stage 23/11–04/12/2026	End-to-end workflows WF-01 … WF-06 and demo scenarios DS-01 … DS-12	100% main scenarios and ≥ 80% exception scenarios pass
Acceptance	Whole team with the supervisor	Defense 1.1	Workflows chosen by the committee	Thresholds of guide section 9.4

Constraints and assumptions

- No hosted staging environment exists yet (Design Brief G3); system tests run locally until staging is deployed.
- GHN is tested on the GHN staging environment; Lazada data of the test shop is masked (phone / address).
- Unit tests mock the MongoDB models; there is no automated end-to-end test on a real replica set yet (Design Brief G4).
- Only the back-end Jest suite (be/src/**/*.spec.ts) and the packer pytest suite run in CI; be/test (e2e) is run on demand.

2. Test Strategy

2.1 Testing Types

Type	Objective	Technique	Completion criteria
Functional	Each feature behaves as specified in SRS section 3	Equivalence classes on inputs; main flow, business rule and exception per feature	Every feature has 3 passing test cases
Business rule / data validation	Rules BR-01 … BR-17 and DTO validation are enforced	Boundary values (15-day return window, 72-hour password, weight tolerance, ordered quantity)	All rule cases pass; error codes match the SRS message list
Security	Authentication, lockout, role checks, signed callbacks	Negative tests (wrong password, wrong role, wrong signature, replayed webhook)	401 / 403 returned in every negative case
State transition	Order Group moves only through allowed transitions	One case per allowed transition; invalid jumps must be rejected	33 transitions covered
Integration	External systems are called and handled correctly	AURELLE simulator and conformance script, GHN staging, idempotency keys	Conformance script passes; duplicates ignored
Regression	Changes do not break existing behaviour	Full Jest and pytest suites in GitHub Actions on every push	CI green before merge to main
Performance	Packing and synchronization meet NFR thresholds	Benchmark scripts on 300 orders; timed test orders	NFR-1 and NFR-2 thresholds met
Reliability / availability	No lost status change; packing works without the packer service	Simulated marketplace errors (retry, back-off); packer service stopped	NFR-4 and NFR-6 met
User interface (manual)	Screens and buttons of each role work as in SRS 3.1.2	Manual run of demo scenarios with seeded data	All main scenarios pass

2.2 Test Levels

Type of Tests	Unit	Integration	System	Acceptance
Functional	X	X	X	X
Business rule / data validation	X	X	X	
Security	X	X	X	X
State transition	X		X	
Integration		X	X	
Regression	X	X		
Performance		X	X	
Reliability / availability	X	X	X	
User interface (manual)			X	X

2.3 Supporting Tools

Purpose	Tool	Vendor/In-house	Version
Back-end unit tests	Jest + ts-jest	Open source	Jest 30, ts-jest 29
HTTP tests (e2e, on demand)	supertest	Open source	7
Packer service tests and checks	pytest, ruff, mypy	Open source	pytest 9.1.1, ruff 0.16.10, mypy 2.4.0
Continuous integration	GitHub Actions (ci.yml)	GitHub	ubuntu-latest, Node.js 22, Python 3.12
AURELLE Open API simulator and conformance	be/scripts/aurelle-mock-server.ts, aurelle-conformance.ts, aurelle-webhook-smoke-test.ts	In-house	tag review-2
Carrier checks	be/scripts/ghn-smoke-test.ts, ghn-masked-order-test.ts	In-house	tag review-2
Packing benchmarks	be/scripts/solver-benchmark.ts, cpsat-benchmark.ts	In-house	tag review-2
API exploration	Swagger UI (/api/docs)	Open source	@nestjs/swagger 11
Test case management	Report5_Test_Report.xlsx, Report5_Unit_Test.xlsx	In-house	1.0

3. Test Plan

3.1 Human Resources

Worker/Doer	Role	Specific Responsibilities/Comments
Nguyễn Phương Mỹ Thuận	Leader, back-end developer, test lead	Test plan and test report; unit tests of auth, marketplace integration and shipping; owner of WF-01 and WF-04 test cases; NFR-1 measurement
Lê Đức Trung Thi	Back-end developer	Unit tests of order groups, packing solver and returns; owner of WF-05 and WF-06 test cases; benchmarks (NFR-2); end-to-end tests (G4)
Huỳnh Quốc Việt	Front-end developer	Manual UI tests of picking, packing, shipping and returns screens; owner of WF-03 test cases
Phan Huỳnh Hải Phượng	Front-end developer (admin)	Manual UI tests of admin, warehouse and marketplace screens; owner of WF-02 test cases

3.2 Test Environment

Purpose	Tool	Provider	Version
CI runner	GitHub Actions ubuntu-latest	GitHub	Node.js 22, Python 3.12
Database	MongoDB Atlas (replica set, transactions)	MongoDB	Atlas cluster
Cache	Redis (Docker container redis:7-alpine)	Redis	7
Back end	NestJS on Node.js	Open source	NestJS 11
Staff portal / AURELLE website	React + Vite / Next.js	Open source	React 19.2, Vite 8 / Next.js 15.5
Packer service	FastAPI + OR-Tools CP-SAT	Open source	Python 3.12, FastAPI 0.142, OR-Tools 9.15
Carrier	GHN staging API	GHN	API v2
Marketplace	Lazada test shop (Open Platform); AURELLE Open API simulator (port 4000)	Lazada / in-house	—
Browser	Google Chrome	Google	latest

3.3 Test Milestones

Milestone Task	Start Date	End Date
Create test cases (Iteration 1)	14/10/2026	20/10/2026
Code, unit test, integration test (Iteration 1)	14/10/2026	23/10/2026
Review 2: test documentation and round 1 (CI)	25/10/2026	25/10/2026
Create test cases (Iteration 2)	28/10/2026	03/11/2026
Code, unit test, integration test (Iteration 2)	28/10/2026	06/11/2026
Create test cases (Iteration 3)	11/11/2026	17/11/2026
Execute system test and fix bugs (Verification)	23/11/2026	04/12/2026
Execute UAT and fix bugs	09/12/2026	11/12/2026

4. Test Cases

Test cases are kept in two workbooks that use the templates of the course:

- **Unit test cases: Report5_Unit_Test.xlsx** – 5 core back-end functions (AuthService.login, OrderGroupsService.pickItem, Order Group transition rules, ReturnsService, packing solver) with 51 unit test cases (normal 20, abnormal 23, boundary 8). Every case points to the Jest test (file and line) that runs it in CI.
- **Other test cases (IT, ST, AT): Report5_Test_Report.xlsx** – 75 functional test cases in 13 sheets: F-01 … F-10, WF (end-to-end), ST (state transitions) and NFR.

Each functional test case has the fields required by the Review 2 guide: ID, related feature or workflow, pre-conditions, steps, test data, expected result, run method (automated or manual) and actual result.

Group	Test case IDs	Count	Automated in CI	Manual
Mandatory features	TC-F01-01 … TC-F10-03	30	30	0
Main workflows	TC-WF01 … TC-WF06	6	0	6
Order Group state transitions	TC-ST01 … TC-ST33	33	27	6
NFRs	TC-NFR1 … TC-NFR6	6	4	2 (NFR-1 timed test, NFR-2 benchmark script)
Total		**75**	**61 (81%)**	**14**

The automated share (81%) meets the threshold of at least 50% (guide section 9.4).

5. Test Reports

Round 1 – 25/10/2026, tag review-2 (commit 1a3101c). Automated cases are taken from GitHub Actions CI run #258 on that commit (all jobs green); the NFR-2 result is the last recorded benchmark run of 04/10/2026 (be/scripts/benchmark-results/solver-v2.json), not re-run on this tag.

Module	Passed	Failed	Pending	Total
F-01 … F-10 (mandatory features)	30	0	0	30
WF (end-to-end workflows)	0	0	6	6
ST (Order Group state transitions)	27	0	6	33
NFR	5	0	1	6
Total	**62**	**0**	**13**	**75**

Test coverage (executed / total) = 82.7%; test successful coverage (passed / total) = 82.7%. Unit test report: 51 of 51 unit cases passed (100%). The whole back-end suite in CI contains 1,251 Jest tests in 121 spec files (status of 23/10/2026, Report 2), plus the packer pytest suite.

Analysis

- No failed test and no open Critical or High defect in round 1.
- The 13 pending cases are manual: the six end-to-end workflows (TC-WF01 … TC-WF06), six Order Group transitions that have no dedicated unit test yet (TC-ST02, TC-ST05, TC-ST08, TC-ST12, TC-ST19, TC-ST24) and the NFR-1 timed synchronization test. They are scheduled for the Verification stage (23/11–04/12/2026), on staging once it is available.
- Planned improvements: automated end-to-end tests for pick → pack → ship → return (Design Brief G4, Thi, 04/12/2026) and unit tests for the six uncovered transitions.